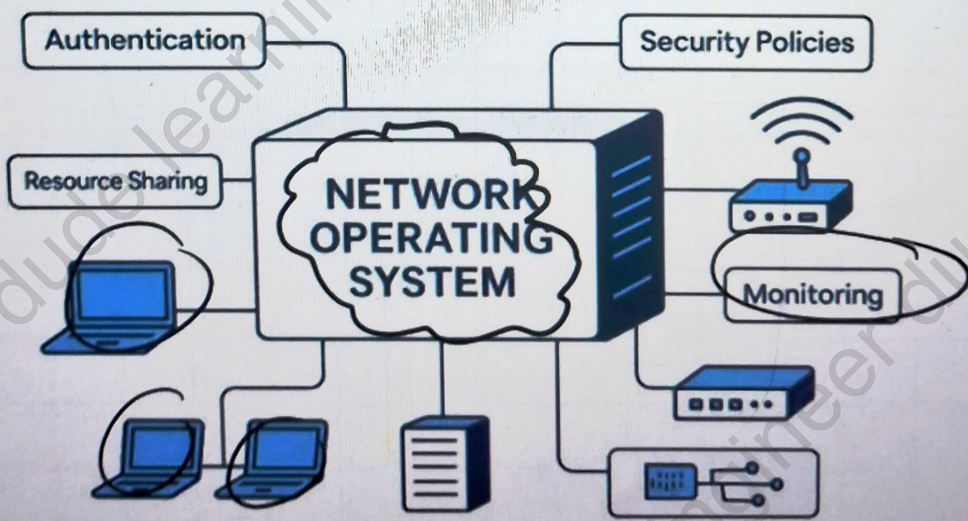
ONESHOT

UNIT-5

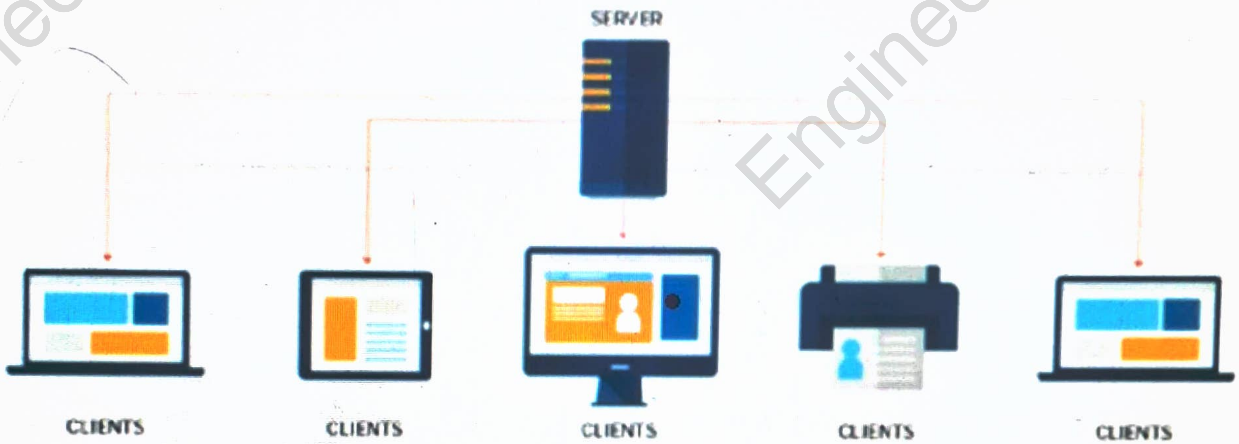
Network Operating System

A **Network Operating System (NOS)** is an operating system that manages computers connected through a network.
Its main purpose is to help computers **communicate and share resources** such as files, printers, applications, and storage.



Need of Network Operating System

Network operating system model



A Network OS is needed to:

- Connect multiple computers in a network
- Share files and folders
- Share printers and other devices
- Manage users and their permissions
- Provide security to network resources
- Allow remote access to resources
- Make network administration easier

Main Features of Network OS

Resource Sharing

Users can share files, printers, storage and other devices.

User Management

The administrator can create users, passwords and access permissions.

File Management

Files can be stored and shared from a central location.

Security

It protects network resources from unauthorized users.

Remote Access

Users can access files and services from another computer.

Network Administration

The administrator can monitor and control network resources.

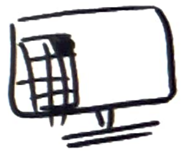
How Network OS Works

The basic working is:

C19

Client Computer → Network → Server → Shared Resources

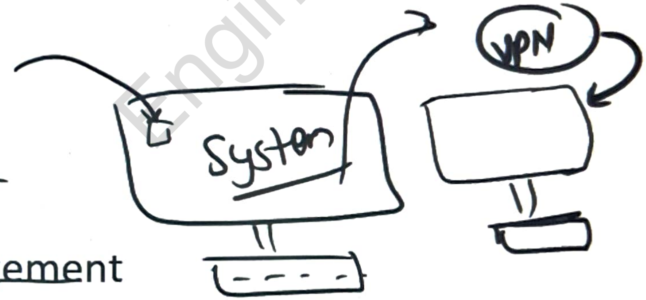
For example, when a user wants to open a shared file:



1. The client sends a request to the server.
2. The server checks the user's permission.
3. If access is allowed, the server sends the requested file.
4. The user can then use the file.

Advantages

- Easy resource sharing
- Centralized user management
- Better security
- Easy backup and storage management
- Remote access
- Easy network administration



Limitations

- Server failure can affect many users
- Requires network setup and maintenance
- Can be more expensive than a simple standalone system
- Requires a skilled administrator

Examples

Common examples include:

- **Windows Server**
- **UNIX**
- **Linux Server**
- **Novell NetWare**

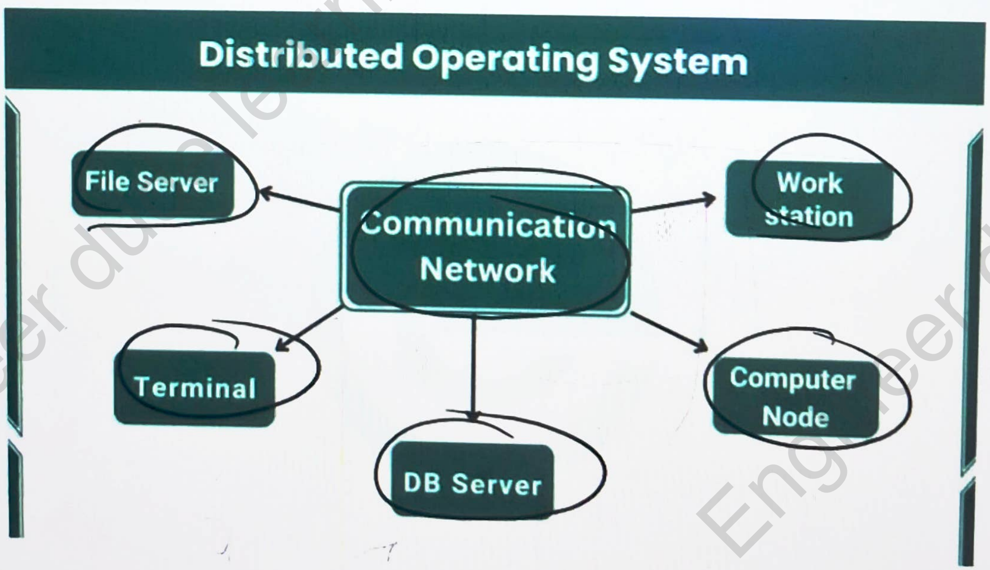
Distributed Operating System

A **Distributed Operating System (DOS)** is an operating system that manages a group of independent computers and makes them work together as a **single system**.

These computers are connected through a network and can share **processing power, memory, files and other resources.**

Simple definition:

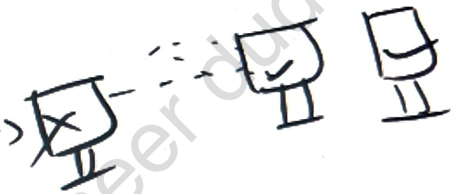
A Distributed OS allows multiple connected computers to work together and complete tasks efficiently.

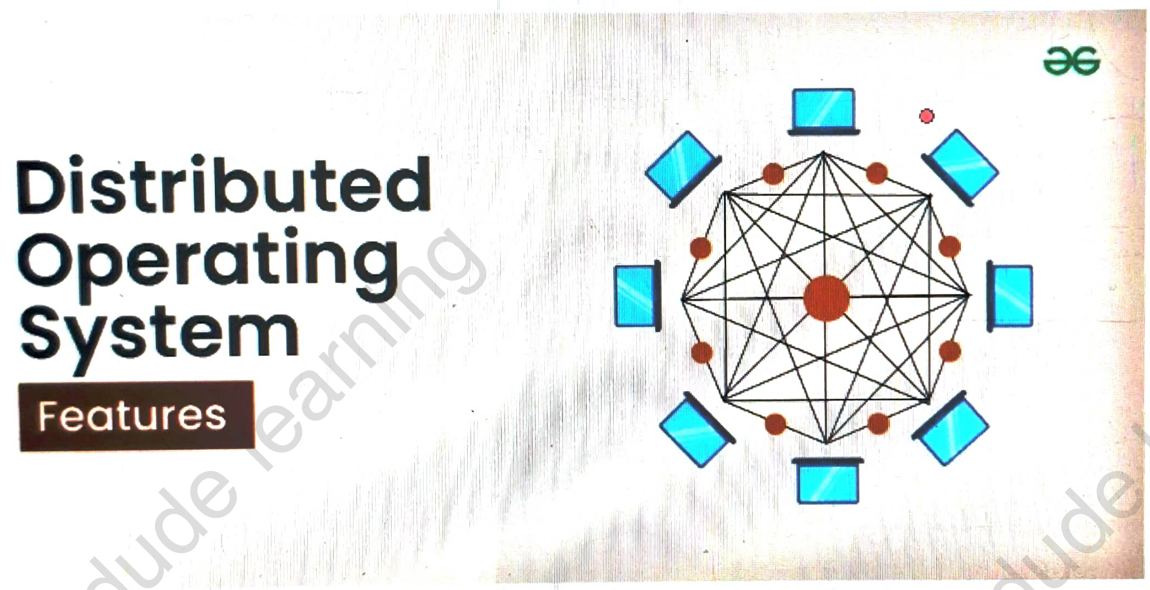


Need of Distributed Operating System

A Distributed OS is needed to:

- Share resources among different computers
- Divide a large task into smaller tasks
- Improve system performance
- Increase reliability
- Use available resources efficiently
- Support communication between computers





Resource Sharing

Different computers can share hardware, software and data.

Transparency

The user does not need to know where a particular resource is located.

Load Sharing

Calculation, (4) parts (4) comput

The workload can be divided among different computers.

Parallel Processing

Multiple computers can work on different parts of a task at the same time.

Reliability

If ~~one computer~~ fails, other computers may continue the work.

Scalability

More computers can be added when required.

Communication

Computers communicate with each other to exchange data and information.

Working of Distributed OS

The basic working is:

User Request → Task Division → Multiple Computers → Result Combination

The Distributed OS divides a large task into smaller parts and sends these parts to different computers.

Each computer processes its assigned part. Finally, the results are collected and combined to produce the final output.

[DIAGRAM]

Make a horizontal diagram:

Large Task

↓

Task Division

↓

1 2 3 4

Computer 1 | Computer 2 | Computer 3 | Computer 4

↓

Combined Result

This should be one of the **main visuals of the topic**.

Advantages of Distributed OS

- Better performance
- Faster processing
- Efficient resource utilization
- High reliability
- Resource sharing
- Can handle large and complex tasks

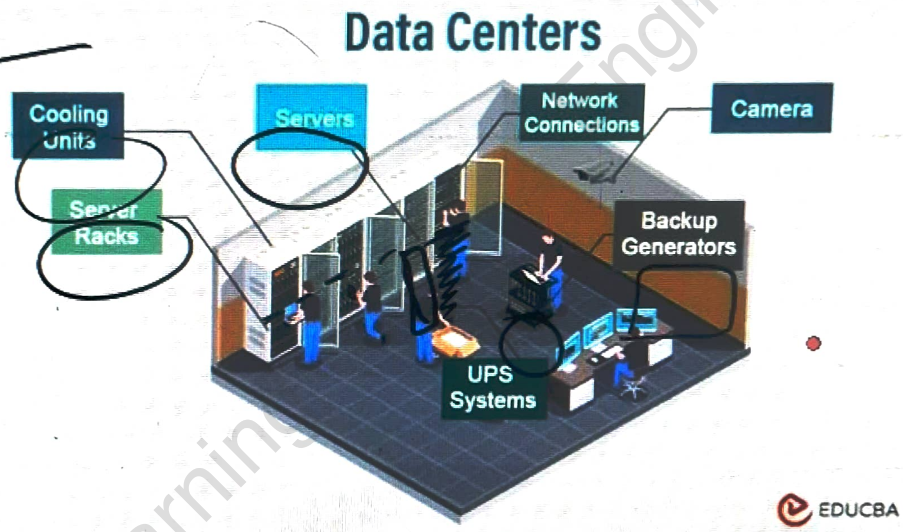
Limitations of Distributed OS

- Network failure can affect communication
- System design is complex
- Security can be difficult
- Proper synchronization is required
- Maintenance can be complicated

Examples

Some examples and distributed-system environments are:

- **Amoeba**
- **Plan 9**
- **LOCUS**
- Distributed systems used in **cloud computing and data CENTER**

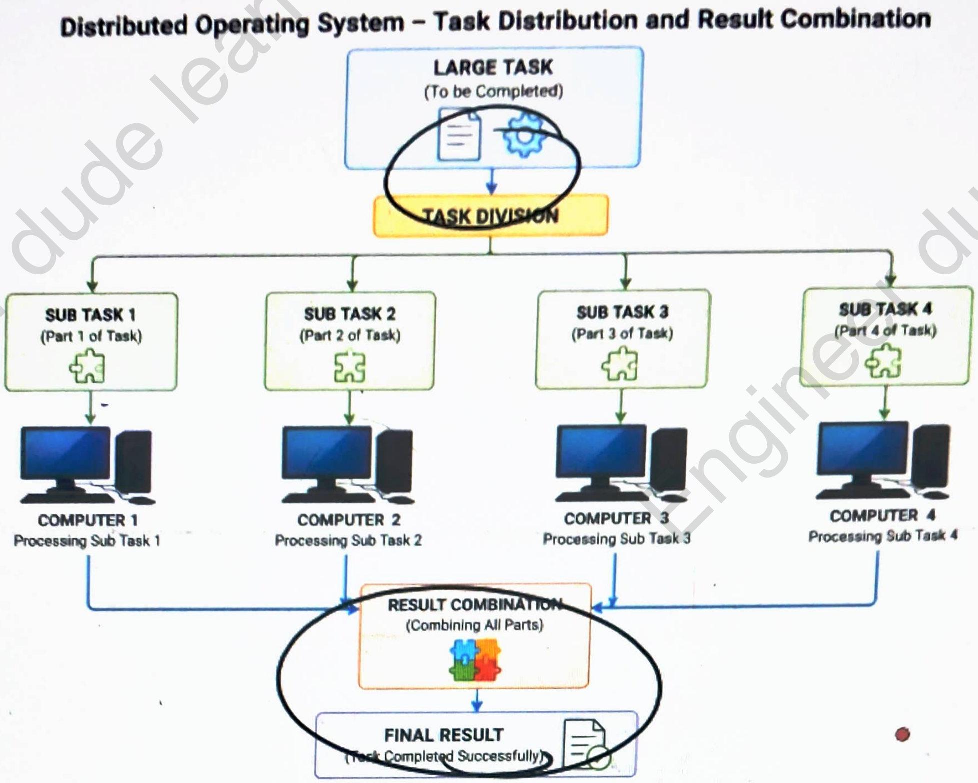


Simple Real-Life Example

Suppose a large task needs to be completed by **4 computers**.
Instead of one computer doing all the work:

Computer 1 → Part 1
Computer 2 → Part 2
Computer 3 → Part 3
Computer 4 → Part 4

The results are then combined to get the final answer.



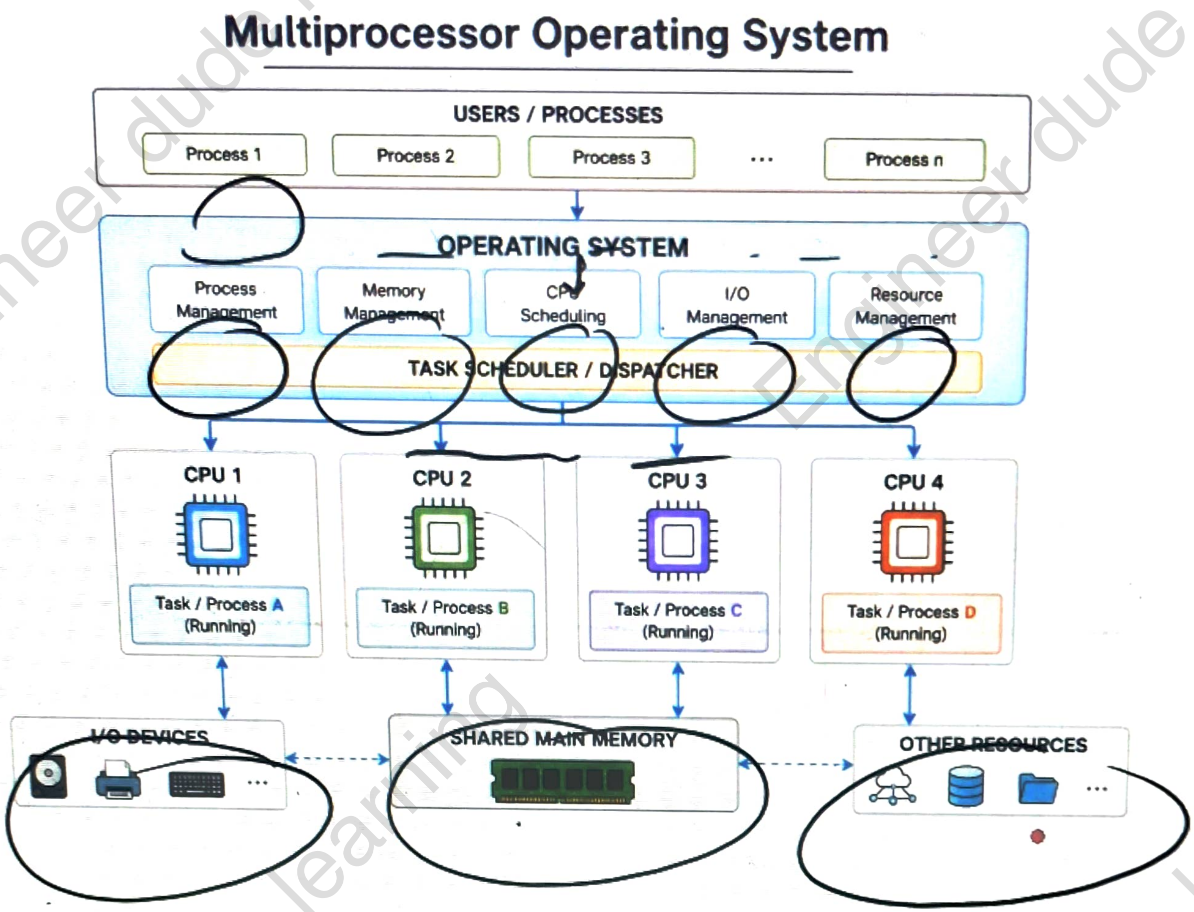
Multiprocessor Operating System

A **Multiprocessor Operating System** is an operating system that manages a computer system having **two or more processors (CPUs)**.

It allows multiple processors to work together and execute tasks efficiently.

Simple definition:

A Multiprocessor OS manages multiple CPUs and distributes work among them.



Need of Multiprocessor OS

- Increase processing speed
- Run multiple tasks at the same time
- Improve system performance
- Share workload between CPUs
- Increase system reliability
- Use CPU resources efficiently

Main Features

Parallel Processing

Multiple processors can execute different tasks at the same time.

Load Sharing

The workload is distributed among available processors.

Resource Sharing

Processors can share memory, I/O devices and other system resources.

Better Performance

More processors can increase the overall processing capacity.

Reliability

If one processor has a problem, other processors may continue running the system, depending on the design.

Multitasking

Many processes can run simultaneously using different processors.

Types of Multiprocessor Systems

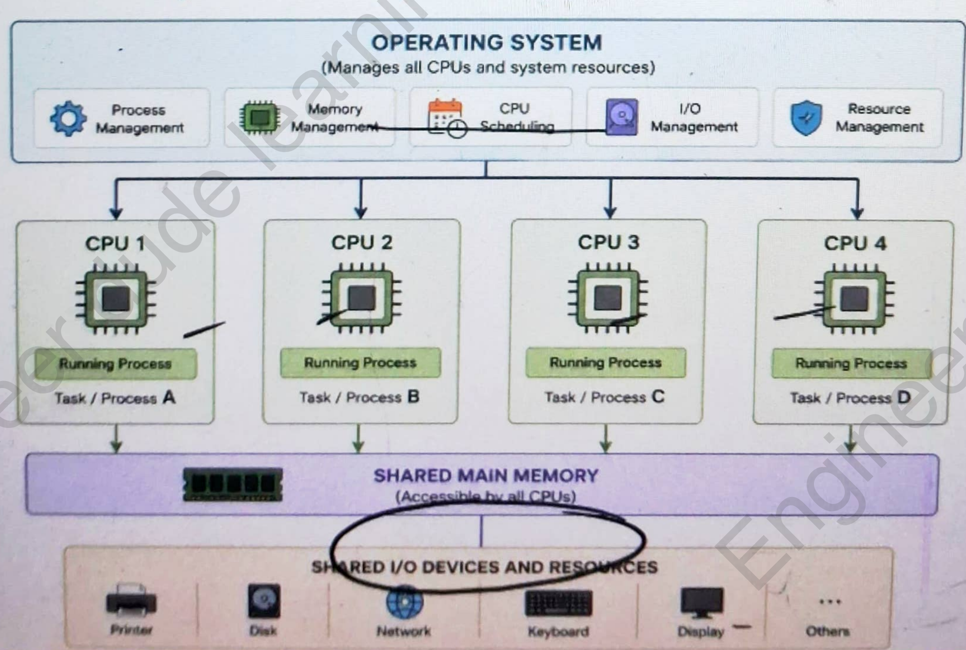
There are mainly two types:

1. Symmetric Multiprocessing (SMP)

In SMP, **all processors are treated equally**.

- Each processor can run processes.
- All processors have access to shared memory.
- Work is distributed among processors.
- There is no single master processor.

Symmetric Multiprocessing (SMP) Architecture

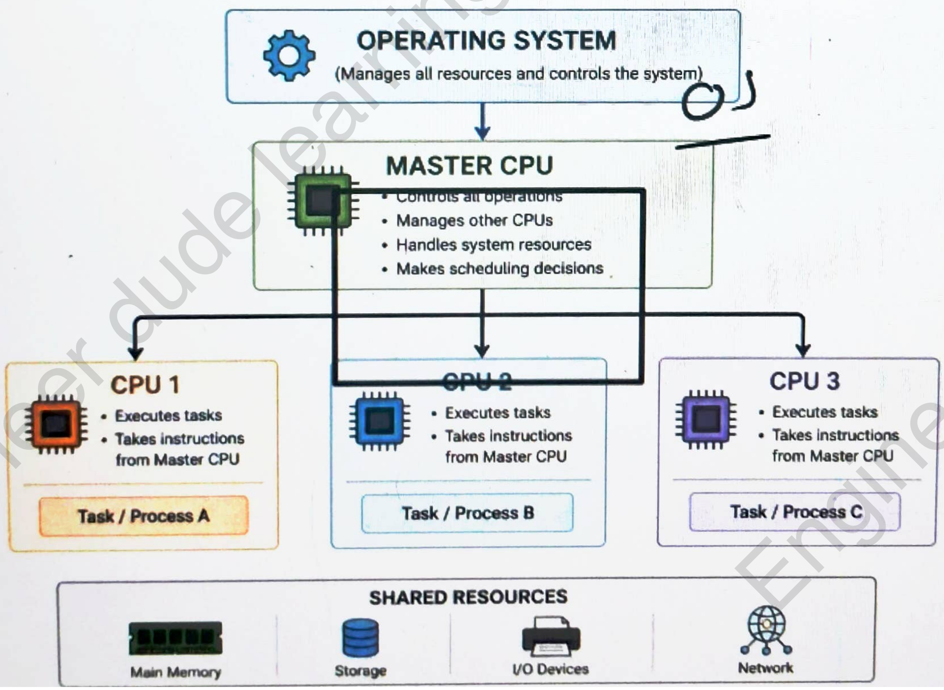


2. Asymmetric Multiprocessing (AMP)

In AMP, one processor acts as the **main/master processor**, while other processors perform assigned tasks.

- Master processor controls the system.
- Other processors perform specific tasks.
- Easier to design than some fully symmetric systems.
- Work is controlled by the master processor.

Asymmetric Multiprocessing (AMP) Architecture



Working of Multiprocessor OS

The operating system receives multiple processes and assigns them to available processors.

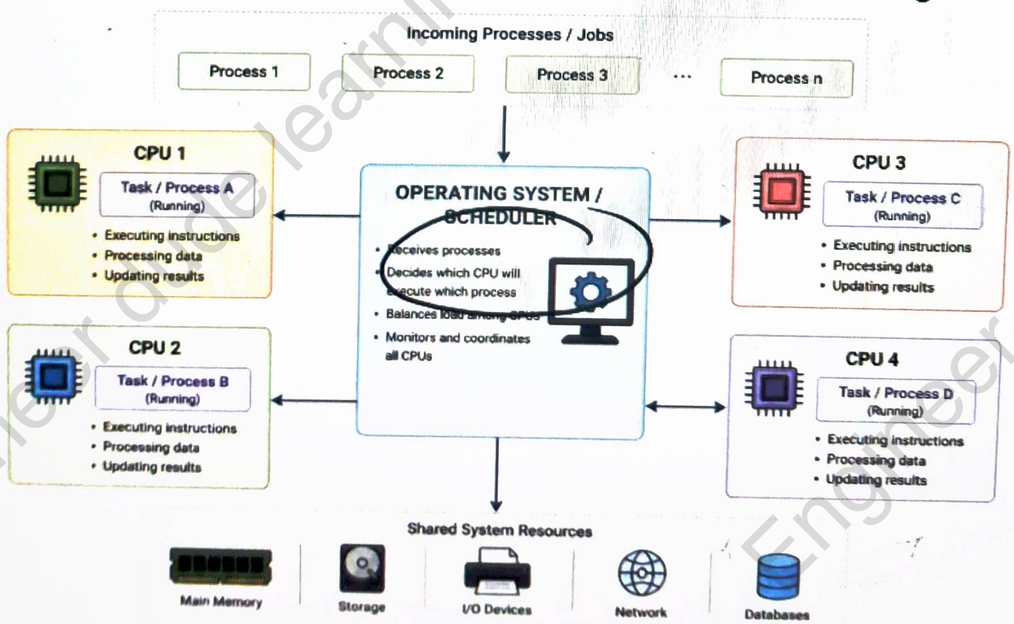
Processes → Scheduler → Multiple CPUs → Results

For example:

- CPU 1 → Process A
- CPU 2 → Process B
- CPU 3 → Process C
- CPU 4 → Process D

All processors work simultaneously and improve the overall performance.

Multiprocessor Operating System – Task Scheduling



Advantages

- Faster processing
- Better CPU utilization
- Supports parallel execution
- Better multitasking
- Improved performance
- Can support large and complex applications

Limitations

- More expensive hardware
- Complex OS design
- Synchronization between processors is required
- More difficult to manage
- Failure of shared resources can affect the system

Applications

Multiprocessor systems are commonly used in:

- Servers
- Data CENTER
- Scientific computing
- Engineering applications
- High-performance computing
- Large databases

Case Study — UNIX / Linux

UNIX

UNIX is a powerful operating system designed for **multiuser and multitasking** environments.
It was originally developed at **AT&T Bell Labs** and became widely used in servers, workstations and academic systems.

Main Characteristics

- Supports multiple users
- Supports multitasking
- Provides strong security
- Uses a hierarchical file system UNIT
- Supports networking
- Provides a command-line interface
- Stable and reliable

Features of UNIX

Multiuser

Many users can use the system at the same time.

Multitasking

A user can run multiple programs simultaneously.

Security

UNIX provides users, groups and file permissions.

Portability

UNIX can run on different types of hardware.

Networking

It ~~provides strong~~ support for computer networks.

Shell

The shell ~~allows users to~~ interact with the operating system using commands.

UNIX Architecture

UNIX mainly has these layers:

Users → Shell → Kernel → Hardware

Shell

UNIX

Acts as an interface between the user and the kernel.

Kernel

The core part of UNIX that manages:

- Processes
- Memory
- Files
- Devices
- Security

Kernel

Hardware

Includes CPU, memory, storage and I/O devices.

USER

↓

SHELL

↓

KERNEL

↓

HARDWARE

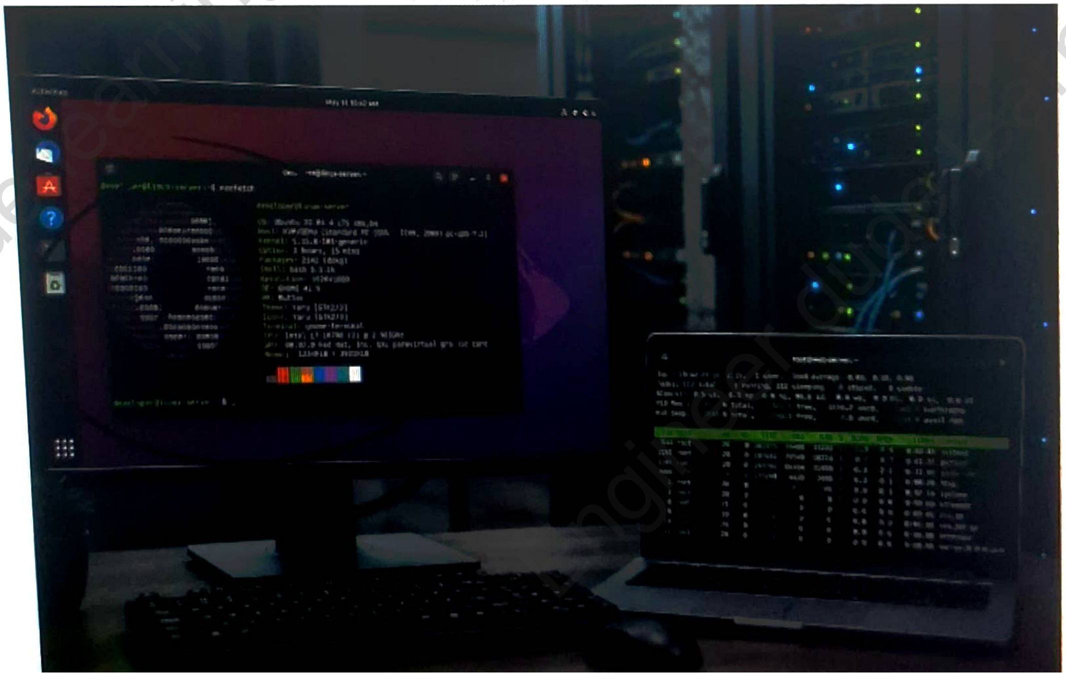
Linux

~~**Linux**~~ is an **open-source operating system** based on UNIX concepts.

It is widely used in:

- ~~Servers~~
- Cloud computing
- Programming
- Supercomputers
- Embedded systems
- Personal computers

~~Linux is~~ available in different **distributions**, each designed for different purposes.



Features of Linux

Open Source

~~Its~~ source code is available for modification and development.

Multiuser

Multiple users can work on the system.

Multitasking

Many programs can run at the same time.

Security

Linux provides permissions, user accounts and other security mechanisms.

Networking

Linux has strong built-in networking support.

Stability

It can run for long periods with good reliability.

Customization

Users can customize the system according to their needs.

Linux Architecture

Linux can be understood using these main layers:

Users / Applications

↓

Shell & System Utilities

↓

Linux Kernel

↓

Hardware

Kernel

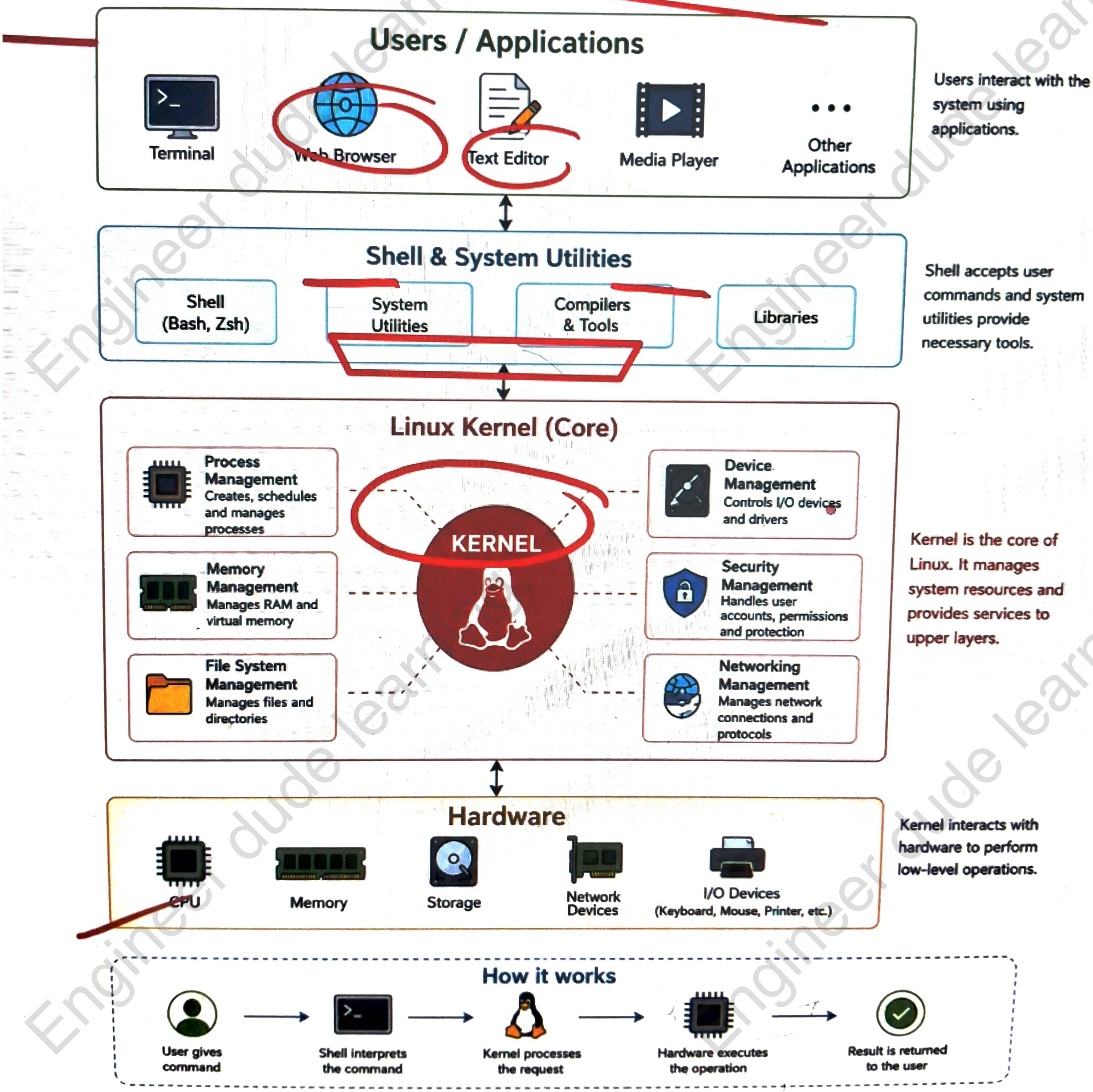
Manages CPU, memory, processes, files and devices.

Shell

Accepts commands from the user and communicates with the kernel.

System Utilities

Provide tools for file management, networking and system administration.


Users / Applications
Terminal
Web Browser
Text Editor
Media Player
Other Applications
Users interact with the system using applications.
Shell & System Utilities
Shell (Bash, Zsh)
System Utilities
Compilers & Tools
Libraries
Shell accepts user commands and system utilities provide necessary tools.
Linux Kernel (Core)
KERNEL
Process Management Creates, schedules and manages processes
Memory Management Manages RAM and virtual memory
File System Management Manages files and directories
Device Management Controls I/O devices and drivers
Security Management Handles user accounts, permissions and protection
Networking Management Manages network connections and protocols
Kernel is the core of Linux. It manages system resources and provides services to upper layers.
Hardware
CPU
Memory
Storage
Network Devices
I/O Devices (Keyboard, Mouse, Printer, etc.)
Kernel interacts with hardware to perform low-level operations.
How it works
User gives command
Shell interprets the command
Kernel processes the request
Hardware executes the operation
Result is returned to the user

Linux Distributions

A **Linux distribution** is a complete operating system built around the Linux kernel.

Popular distributions

- **Ubuntu**
- **Fedora**
- **Debian**
- **Linux Mint**
- **Arch Linux**

Different distributions provide different software, tools and user interfaces.

Popular Linux Distributions

Different Linux distributions for different needs

Ubuntu

User-friendly and very popular.
Great for beginners and general use.

Fedora

Modern, fast and innovative.
Preferred by developers.

Linux

The Kernel

Debian

Stable, secure and reliable.
Widely used on servers.

Linux Mint

Easy to use and beautiful.
Perfect for desktop users.

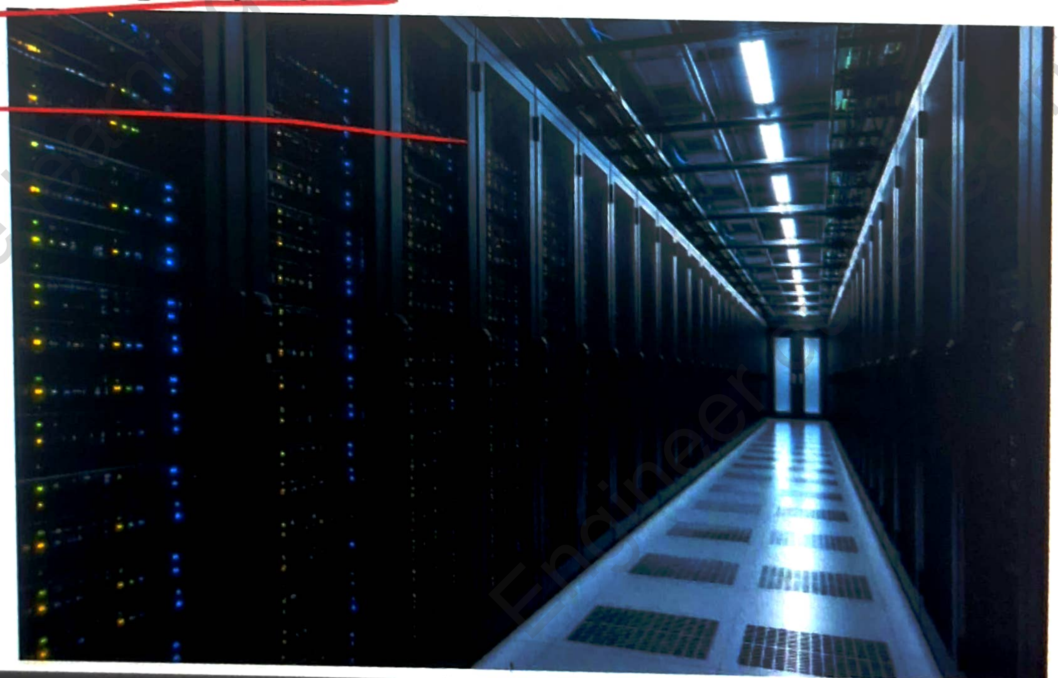
Arch Linux

Highly customizable and lightweight.
For advanced users

Uses of Linux

Linux is commonly used in:

- Web servers
- Cloud platforms
- Data CENTER
- Software development
- Supercomputers
- Embedded devices
- Networking equipment



UNIX vs LINUX

UNIX	Linux
Developed earlier	Developed later
Many UNIX versions are proprietary	Linux is open source
Common in enterprise systems	Common in servers, cloud ar development
Has different commercial versions	Has many free distributions
Strong multiuser and multitasking support	Strong multiuser and multitasking support

Uses of Linux

1 Linux in Web Servers

Linux is widely used to run **web servers** because it is stable, secure and reliable.

Examples: Websites, web applications and online services.

2 Linux in Cloud Computing

Linux is widely used in cloud platforms to manage virtual machines, servers and applications.

It is useful because:

- It is flexible
- It supports virtualization
- It can handle large workloads
- It is easy to customize

3. Linux in Software Development

Developers use Linux for:

- Programming
- Testing applications
- Running development tools
- Using compilers and interpreters
- Managing code through terminal commands

4. Linux in Supercomputers

Linux is widely used in **high-performance and supercomputing systems**.

It helps these systems:

- Process large amounts of data
- Run scientific calculations
- Perform complex simulations
- Handle parallel computing

5. Linux in Embedded Systems

Linux can be used in devices such as:

- Smart TVs
- Routers
- Network devices
- Industrial machines
- IoT devices

6. Linux in Networking

Linux is also used in:

- Network servers
- Routers
- Firewalls
- Network monitoring systems
- Communication systems

Because Linux provides strong networking tools and controls.

Case Study — Windows Operating System

Windows

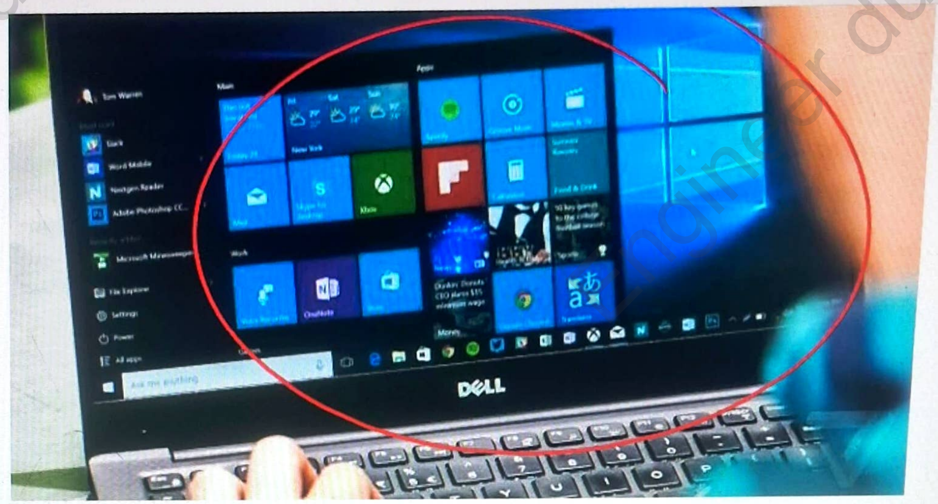
Windows is a graphical operating system developed by **Microsoft**.

It is widely used on personal computers, laptops and business systems.

Windows provides an easy interface through which users can run applications, manage files, connect to networks and use hardware devices.

Main Characteristics

- Graphical User Interface (GUI)
- Multitasking
- File and folder management
- User account management
- Networking support
- Security features
- Hardware support



Features of Windows

Graphical User Interface

Windows provides windows, icons, menus and a mouse-based interface.

Multitasking

Users can run multiple applications at the same time.

File Management

Users can easily create, copy, move, rename and delete files and folders.

...

Hardware Support

Windows supports many devices such as printers, keyboards, storage devices and network hardware.

Networking

It provides tools for connecting computers and sharing resources over a network.

Security

Windows provides user accounts, permissions, firewall and other security features.

Windows Architecture

Windows can be understood through two main parts:

User Mode → Kernel Mode → Hardware

User Mode

Applications and user programs run in this mode.

Examples:

- Web browsers
- Office applications
- Media players

Kernel Mode

This is the core part of Windows.

It manages:

- CPU
- Memory
- Processes
- Devices
- File system
- System security

Hardware

Includes processor, RAM, storage and I/O devices.

Applications

↓

User Mode

↓

Windows Executive + Kernel

↓

Device Drivers

↓

Hardware

Engineer dude learning

File Management in Windows

Windows uses **files and folders** to organize data.

Users can:

- Create files and folders
- Copy and move files
- Rename files
- Delete files
- Search for files
- Set file permissions

File Explorer

File Explorer is the main tool used to browse and manage files and folders.

Engineer dude learning

User Management

Windows allows administrators to create and manage user accounts.

Main Functions

- Create user accounts
- Set passwords
- Assign permissions
- Manage administrators and standard users
- Control access to files and resources

Types of Users

Administrator

Has high-level control over the system.

Standard User

Can perform normal tasks but has limited system privileges.

Networking in Windows

Windows provides support for computer networking.

It allows users to:

- Connect to Wi-Fi and networks
- Share files and folders
- Share printers
- Access network resources
- Communicate with other computers

Windows Server

Windows Server is designed for organizations and network management.

It can provide:

- User management
- File sharing
- Network services
- Application services

Security in Windows

Windows includes several security mechanisms.

User Authentication

Passwords and other login methods help protect accounts.

Permissions

Access to files and resources can be controlled.

Windows Defender

Provides protection against malware and other threats.

Firewall

Helps control unwanted network traffic.

Updates

Security updates improve protection.